Cone Matching and Integral Constraints in Relative Langlands Duality

PIATRA . INSTITUTE

October 2026

Abstract

Chen and Venkatesh proved that two pairs of singular spaces, built from the nilpotent cone of 3×3 matrices and from $(2,2,2)$-tensors of vanishing hyperdeterminant, are numerically weakly dual in the sense of relative Langlands duality. The decisive step of their proof is that the inequalities describing integral points of one space coincide with the inequalities describing the weights of regular functions on the other. They attribute this to coincidences specific to their groups and expect the underlying convex reduction to occur only for groups isogenous to their examples. We analyse the same comparison for every split reductive group with one-dimensional centre. The automorphic inequalities reduce in this way for the root types A_2, B_2, G_2 and A_1^3 and for no others. The two cones coincide exactly when, in addition, the integer $d = \langle \theta_G, \nu_G \rangle$ pairing the primitive central character with the primitive central cocharacter equals $1/(\kappa - 1)$, where $\kappa = \langle \rho, \check{\rho} \rangle$; this leaves A_2 with $d = 1$ and A_1^3 with $d = 2$. Equal cones need not index equal sets of representations. The description of the coordinate ring that the comparison uses requires the extra condition $\langle \lambda, \check{\rho}' \rangle \in \mathbb{Z}$ when the shifted cocharacter $\check{\rho}'$ is only rational, and with it the two supports agree if and only if the shifted cocharacter of the dual group is integral. Of the 35 labelled global forms of type A_1^3 with $d = 2$, five pass the test for both identities: the published pair and three copies of the self-dual group $\mathrm{SO}_4 \times \mathrm{GL}_2$. Exact computer checks cover all root types up to rank 8. No new period identity is proved.

1. Introduction

Relative Langlands duality, proposed by Ben-Zvi, Sakellaridis and Venkatesh, pairs a Hamiltonian space for a reductive group G with a Hamiltonian space for its dual group $\check{G}$, so that periods of automorphic forms on one side correspond to L-functions attached to the other (Ben-Zvi et al. 2024). The proposal extends the program of Sakellaridis and Venkatesh on periods of spherical varieties (Sakellaridis and Venkatesh 2017; Beuzart-Plessis 2025) and is modelled on the electric–magnetic duality of boundary conditions in four-dimensional supersymmetric Yang–Mills theory (Gaiotto and Witten 2009; Kapustin and Witten 2007). Its ingredients include the dual groups attached to spherical varieties (Sakellaridis and Venkatesh 2017; Knop and Schalke 2017), and categorical forms of its conjectures have been tested in low-rank cases (Feng and Wang 2025). In its original form both spaces are smooth. Chen and Venkatesh studied what happens when they are not (Chen and Venkatesh 2026). Their Theorem 7.1 shows that two pairs of singular spaces satisfy a numerical form of the duality: for $G = \mathrm{PGL}_3 \times \mathbb{G}_m$ the space X is the nilpotent cone of 3×3 matrices, and one of the two identities is essentially Ginzburg's integral representation of the adjoint L-function of GL_3 (Ginzburg 1991); for a group isogenous to $\mathrm{SL}_2^3 \times \mathbb{G}_m$ the space X consists of $(2,2,2)$-tensors of vanishing hyperdeterminant, and one identity is essentially Garrett's integral representation of the triple product L-function (Garrett 1987). In both cases the other identity is new and does not express a quotient of standard L-values. For singular spherical varieties an established approach replaces the characteristic function of integral points by an exotic basic function built from intersection cohomology of arc spaces, whose local factors recover L-values (Braverman and Kazhdan 1999; Sakellaridis 2012; Bouthier et al. 2016; Sakellaridis and Wang 2022). Chen and Venkatesh

keep the naive basic function and find that the theory remains dual, with a dual space that is itself singular (Chen and Venkatesh 2026, sec. 1.4).

Both examples have the form $X = \overline{T'U'\backslash G}$, the affine closure of a quotient of G by a one-dimensional torus T' and the kernel U' of a generic character of a maximal unipotent subgroup, and similarly on the dual side. The authors stress that the duality of these spaces is not a general fact: it uses numerical coincidences specific to the groups considered (Chen and Venkatesh 2026, sec. 1.1). The key point of the proof is that the inequalities defining integrality of points of X match the set of weights occurring in functions on $\check{X}$, and vice versa (Chen and Venkatesh 2026, sec. 7). In the toric case this matching holds by construction, because integral points of a toric variety and monomials on its dual are indexed by the same cone (Chen 2024, sec. 1.2 and Definition 2.3). In the singular examples the two sets of inequalities come from different calculations, one about integral points and one about invariant vectors, and they agree only after a convex reduction: one family of automorphic inequalities lies in the convex hull of another. Remark 7.4 of Chen and Venkatesh (2026) records the expectation that "this phenomenon only occurs in cases isogenous to the Ginzburg and Garrett examples."

This note answers two questions about that comparison for an arbitrary split reductive group G with one-dimensional centre. For which root data do the two cones of the comparison coincide? And when they coincide, do the lattice points that index the two local factors coincide as well? The answers separate three inputs that the published examples satisfy simultaneously: the root system, an integer recording how the centre meets the derived group, and the character lattice.

The results are as follows. The convex reduction of the automorphic inequalities occurs exactly for the root types A_2, B_2, G_2 and A_1^3 (Theorem 3). The cones coincide exactly when, in addition, the primitive central pairing d equals $1/(\kappa - 1)$, which leaves $(A_2, 1)$ and $(A_1^3, 2)$ (Corollary 4). The literal reading of Remark 7.4 therefore fails for B_2 and G_2, while the full matching does single out the root types of the published examples. The description of $k[X]$ in Proposition 5.1 of Chen and Venkatesh (2026) needs an integrality condition when the shifted cocharacter $\check{\rho}'$ is rational (Proposition 5). With that condition, equality of cones gives equality of supports if and only if $\check{\rho}'$ is a cocharacter of the dual group (Theorem 6). For root type A_1^3 with $d = 2$ this happens for five of the 35 labelled global forms (Proposition 7). For $\mathrm{GL}_2 \times \mathrm{PGL}_2 \times \mathrm{PGL}_2$, which has the same cones as Garrett's example, one of the two support comparisons fails from central degree 3 onward.

The published period identities are unaffected: in all four groups of Theorem 7.1 of Chen and Venkatesh (2026) the shifted cocharacter is a genuine cocharacter, a point the authors make explicitly for the dual Garrett group. The note proves no new period identity. It concerns the step of the proof that compares supports, and leaves the remaining steps (the orbit structure of X, the integral models, the unfolding and the global constants) as they are in the source.

2. The two cones

2.1 Groups and the central pairing

Let k be a field of characteristic zero and G a split connected reductive group over k whose connected centre $Z(G)^0$ has dimension one. Fix a split maximal torus T in a Borel subgroup $B = TU$, with simple roots Δ and semisimple rank $r = |\Delta|$. Following (Chen and Venkatesh 2026, sec. 5.1), let $\theta_G \colon G/[G,G] \to \mathbb{G}_m$ be an isomorphism, regarded as a character of G and of T, and let $\check{\theta}_G$ be the rational central cocharacter with $\langle \theta_G, \check{\theta}_G \rangle = 1$. Let ρ and $\check{\rho}$ be half the sums of the positive roots and coroots, so that $\langle \alpha, \check{\rho} \rangle = \langle \rho, \check{\alpha} \rangle = 1$ for every simple root α (Humphreys 1972, sec. 10.2), and put $\check{\rho}' = \check{\rho} + \check{\theta}_G$ and $T' = \operatorname{im} \check{\rho}'$. Both $\check{\theta}_G$ and $\check{\rho}'$ may be defined only on a finite cover of $\mathbb{G}_m$; the source allows this explicitly. Let U' be the kernel of the homomorphism $U \to \mathbb{G}_a$ that is the identity on every pinned simple root group, $S = T'U'$, and $X = \overline{S \backslash G}$ the affine variety with coordinate ring $k[G]^S$. The Langlands dual group $\check{G}$ has the dual root datum (Springer 1998; Borel 1979), and $\check{X}$ is built from $\check{G}$ in the same way.

Let $\nu_G \colon \mathbb{G}_m \to Z(G)^0$ be the isomorphism for which

$$d = d_G := \langle \theta_G, \nu_G \rangle$$

is positive. We call d the primitive central pairing.

Lemma 1. (a) d is a positive integer, equal to the order of $Z(G)^0 \cap [G,G]$. (b) $\check{\theta}_G = \nu_G/d$. (c) Under $X^*(T) = X_*(\check{T})$, the character θ_G is the primitive central cocharacter of $\check{G}$, and $\theta_G = d\,\check{\theta}_{\check{G}}$; dually $\nu_G = \theta_{\check{G}}$ under $X_*(T) = X^*(\check{T})$. In particular $d_{\check{G}} = d_G$.

Proof. (a) The pairing of a character and a cocharacter is an integer, and $\theta_G \circ \nu_G(t) = t^d$. The kernel of θ_G is $[G,G]$, so θ_G restricted to $Z(G)^0 \cong \mathbb{G}_m$ has kernel $Z(G)^0 \cap [G,G]$, which is the group μ_d of order d. (b) The rational central cocharacters form the line $\mathbb{Q}\nu_G$, and $\langle \theta_G, \nu_G/d \rangle = 1$. (c) Characters of G are the characters of T orthogonal to all coroots, and central cocharacters of $\check{G}$ are the cocharacters of $\check{T}$ orthogonal to all roots of $\check{G}$, which are the coroots of G. The two lattices are equal and of rank one, and θ_G generates the first, so it generates the second. By the same argument $\theta_{\check{G}} = \nu_G$ with the signs chosen to make pairings positive. Then $d_{\check{G}} = \langle \theta_{\check{G}}, \nu_{\check{G}} \rangle = \langle \nu_G, \theta_G \rangle = d_G$, and (b) applied to $\check{G}$ gives $\check{\theta}_{\check{G}} = \nu_{\check{G}}/d = \theta_G/d$. □

For $\mathrm{PGL}_3 \times \mathbb{G}_m$ and $\mathrm{SL}_3 \times \mathbb{G}_m$, with θ_G the projection to the second factor, $d = 1$. For GL_3, with $\theta_G = \det$ and $\nu_G(t) = tI_3$, $d = 3$. For GL_2, $d = 2$, and $d = 2$ for both groups of Garrett's example (Chen and Venkatesh 2026, (51), (53)). The relation $\theta_G \leftrightarrow \gamma\,\check{\theta}_{\check{G}}$ in (57) of Chen and Venkatesh (2026) is Lemma 1(c) with $\gamma = d$; in the source, γ is also defined as $1/(\kappa - 1)$ (Chen and Venkatesh 2026, (56)), and the coincidence of these two numbers is the first of the conditions found below.

2.2 Coordinates

Write $C_{\alpha\beta} = \langle \alpha, \check{\beta} \rangle$ for the Cartan matrix and $c_{\alpha\beta}$ for the entries of C^{-1}. Let $\varpi_\alpha \in X^*(T)_{\mathbb{Q}}$ be the weight dual to the simple coroots and trivial on central cocharacters, so $\varpi_\alpha = \sum_\beta c_{\alpha\beta}\beta$. Then

$$\rho = \sum_\beta n_\beta\, \beta, \quad n_\beta = \sum_\alpha c_{\alpha\beta}, \qquad \check{\rho} = \sum_\alpha m_\alpha\, \check{\alpha}, \quad m_\alpha = \sum_\beta c_{\alpha\beta}, \qquad \kappa = \langle \rho, \check{\rho} \rangle = \sum_\beta n_\beta = \sum_\alpha m_\alpha.$$

The source writes a (arXiv version) or κ (journal version) for $\langle \rho, \check{\rho} \rangle$ and $n_{\check{\alpha}}$ for m_α. A real cocharacter $\chi \in X_*(T)_{\mathbb{R}}$ is determined by the numbers

$$x_\alpha = \langle \chi, \alpha \rangle \ (\alpha \in \Delta), \qquad z = \langle \chi, \theta_G \rangle,$$

because the simple roots and θ_G form a basis of $X^*(T)_{\mathbb{R}}$. In these coordinates

$$R(x) := \langle \chi, \rho \rangle = \sum_\beta n_\beta x_\beta, \qquad \langle \chi, \rho - \alpha \rangle = R(x) - x_\alpha, \qquad W_\alpha(x) := \frac{\langle \chi, \varpi_\alpha \rangle}{m_\alpha} = \frac{\sum_\beta c_{\alpha\beta} x_\beta}{\sum_\beta c_{\alpha\beta}}.$$

Every $c_{\alpha\beta}$ is nonnegative, so W_α is a weighted mean of the x_β.

2.3 The automorphic cone

Proposition 5.5 of Chen and Venkatesh (2026) determines which points of $X(F_v)$ of the form $Su_t\varpi^\chi$ are integral, for any G as above with $\kappa > 1$. In the computation of §7.3, combined with the vanishing of the integral against the additive character outside a range of valuations and after the reindexing of §7.3.1, it shows that the local factor of the automorphic X-period is a sum of Weyl characters $s_\chi(\mathrm{Sat}_v)$, weighted by $q_v^{-\langle \chi, \theta_G \rangle/2}$, over the dominant $\chi \in X_*(T)$ satisfying (Chen and Venkatesh 2026, (66) and Proposition 7.3)

$$\frac{\langle \chi, \rho \rangle}{\kappa - 1} \ \geq \ \langle \chi, \theta_G \rangle \ \geq \ \max_{\alpha \in \Delta} \left(\frac{\langle \chi, \rho - \alpha \rangle}{\kappa - 1}, \ \frac{\langle \chi, \varpi_\alpha \rangle}{m_\alpha} \right).$$

The source carries out this computation for the groups of its Theorem 7.1, but the inequalities come from Proposition 5.5 and the conductor bound (64), which hold for every G as above with $\kappa > 1$, and we use them to define the comparison in general. We call the set of real χ satisfying them with $x \geq 0$ the automorphic cone

$$\mathcal{A} = \left\{ x \geq 0, \ \frac{R}{\kappa - 1} \ \geq \ z \ \geq \ \max\left(\max_\alpha \frac{R - x_\alpha}{\kappa - 1}, \ \max_\alpha W_\alpha \right) \right\}.$$

Dropping the inequalities $z \geq W_\alpha$ gives the simplified cone

$$\mathcal{B} = \left\{ x \geq 0, \ \frac{R}{\kappa - 1} \ \geq \ z \ \geq \ \max_\alpha \frac{R - x_\alpha}{\kappa - 1} \right\},$$

which is the form (58) that the automorphic inequalities take in the published examples, where the inequalities $z \geq W_\alpha$ follow from the others because each ϖ_α/m_α lies in the convex hull of the points $(\rho-\beta)/(\kappa-1)$ (Chen and Venkatesh 2026, sec. 7.1(2)). Since $R - x_\alpha \leq R$ for $x \geq 0$, the range of z in $\mathcal{B}$ is nonempty for every dominant x. Always $\mathcal{A} \subseteq \mathcal{B}$.

2.4 The spectral cone

On the spectral side, Proposition 5.1 of Chen and Venkatesh (2026) states that $k[\check{X}]$ is the direct sum of the duals V_λ^* of the irreducible representations of $\check{G}$ whose highest weights $\lambda \in X^*(\check{T})$ satisfy $0 \leq \langle \lambda, \check{\rho}'_{\check{G}} \rangle \leq \min_{\check{\alpha}} \langle \lambda, \check{\alpha} \rangle$. Replacing λ by $-w_0\lambda$ turns this into the statement that the representations V_χ in $k[\check{X}]$ have highest weights χ with $\langle \chi, \check{\rho} \rangle \geq \langle \chi, \check{\theta}_{\check{G}} \rangle \geq \max_\alpha \langle \check{\rho} - \check{\alpha}, \chi \rangle$, in the notation of $\check{G}$ (Chen and Venkatesh 2026, (69)–(70)). Under $X^*(\check{T}) = X_*(T)$ the element $\check{\rho}$ of $\check{G}$ is ρ, the simple coroots of $\check{G}$ are the simple roots of G, and $\check{\theta}_{\check{G}} = \theta_G/d$ by Lemma 1. In the coordinates of §2.2 the spectral cone is therefore

$$\mathcal{S}_d = \left\{ R \;\geq\; \frac{z}{d} \;\geq\; R - \min_\alpha x_\alpha \right\}.$$

Its inequalities force $\min_\alpha x_\alpha \geq R - z/d \geq 0$, so $\mathcal{S}_d$ lies in the dominant chamber. In the published examples $d = 1/(\kappa-1)$, and the source concludes from the convex reduction that the two descriptions are "exactly equivalent" (Chen and Venkatesh 2026, sec. 7.4).

Lemma 2. $\mathcal{A} = \mathcal{S}_d$ if and only if $d(\kappa-1) = 1$ and $\mathcal{A} = \mathcal{B}$.

Proof. On the line $x_\alpha = 1$ for all α we have $R = \kappa$, $(R - x_\alpha)/(\kappa-1) = 1$ and $W_\alpha = 1$. The cone $\mathcal{A}$ meets this line in $1 \leq z \leq \kappa/(\kappa-1)$ and $\mathcal{S}_d$ meets it in $d(\kappa-1) \leq z \leq d\kappa$, so equality of the cones forces $d(\kappa-1) = 1$. When $d(\kappa-1) = 1$, dividing the inequalities of $\mathcal{S}_d$ by $\kappa - 1$ gives $\mathcal{B}$, so $\mathcal{A} = \mathcal{S}_d$ if and only if $\mathcal{A} = \mathcal{B}$. □

The two conditions are independent. The first concerns the global group, through d, and the second concerns only the root system.

3. Which cones match

3.1 The convex reduction

Theorem 3. Let Φ be a reduced root system with $\kappa > 1$. Then $\mathcal{A} = \mathcal{B}$ if and only if Φ has type A_2, B_2, G_2 or A_1^3. In these four cases every ϖ_α/m_α is a convex combination $\sum_\beta b_{\beta\alpha}(\rho-\beta)/(\kappa-1)$ with

$$b_{\beta\alpha} = n_\beta - (\kappa-1)\frac{c_{\alpha\beta}}{m_\alpha},$$

given (rows β, columns α) by the matrices in Table 1.

Proof. The cones differ only by the inequalities $z \geq W_\alpha$, and for each dominant x the admissible z in $\mathcal{B}$ form a nonempty interval with lower end $\max_\beta (R - x_\beta)/(\kappa - 1)$. Hence $\mathcal{A} = \mathcal{B}$ if and only if

$$W_\alpha(x) \;\leq\; \max_\beta \frac{R(x) - x_\beta}{\kappa - 1} \qquad \text{for all } \alpha \text{ and all } x \geq 0. \tag{1}$$

Sufficiency. For any α the coefficients $b_{\beta\alpha}$ sum to $\kappa - (\kappa - 1)m_\alpha/m_\alpha = 1$, and

$$\sum_\beta b_{\beta\alpha} \frac{R - x_\beta}{\kappa - 1} = \frac{R - \sum_\beta n_\beta x_\beta + (\kappa - 1)\sum_\beta c_{\alpha\beta} x_\beta / m_\alpha}{\kappa - 1} = W_\alpha$$

identically in x. When all $b_{\beta\alpha}$ are nonnegative, W_α is a convex combination of the numbers $(R - x_\beta)/(\kappa - 1)$ and (1) holds. Table 1 shows that this is the case for the four listed types.

Necessity. Test (1) at the dominant point $x = e_\gamma$ with $x_\gamma = 1$ and all other coordinates zero. Then $W_\gamma = c_{\gamma\gamma}/m_\gamma$ and $R = n_\gamma$; since $\kappa > 1$ forces $r \geq 2$, the maximum on the right of (1) is $n_\gamma/(\kappa - 1)$, attained at any $\beta \neq \gamma$. So (1) implies

$$\kappa - 1 \;\leq\; q_\gamma := \frac{n_\gamma m_\gamma}{c_{\gamma\gamma}} \qquad \text{for every simple root } \gamma. \tag{2}$$

The number q_γ depends only on the irreducible component containing γ, because C is block diagonal, while κ is the sum of the values of κ over the components. Take γ to be the first node in Bourbaki's numbering (Bourbaki [1968] 2002, Plates I–IX). For A_ℓ one has $n_1 = m_1 = \ell/2$, $c_{11} = \ell/(\ell+1)$ and $\kappa = \ell(\ell+1)(\ell+2)/12$, so $q_1 = \ell(\ell+1)/4$ and $12(\kappa - 1 - q_1) = (\ell-1)\ell(\ell+1) - 12$, which is positive for $\ell \geq 3$. For B_ℓ and C_ℓ, writing $\varpi_1 = \varepsilon_1$, ρ and $\check{\rho}$ in the standard coordinates gives $c_{11} = 1$, $\{n_1, m_1\} = \{\ell, \ell - \frac{1}{2}\}$ and $\kappa = \ell(\ell+1)(4\ell-1)/12$, so $q_1 = \ell(2\ell-1)/2$ and $12(\kappa - 1 - q_1) = \ell(\ell-1)(4\ell-5) - 12$, positive for $\ell \geq 3$. For D_ℓ with $\ell \geq 4$, $c_{11} = 1$, $n_1 = m_1 = \ell - 1$ and $\kappa = \ell(\ell-1)(2\ell-1)/6$, so $6(\kappa - 1 - q_1) = (\ell-1)(\ell-2)(2\ell-3) - 6 > 0$. For the exceptional types an exact computation gives $(q_1, \kappa - 1) = (48, 77)$ for E_6, $(289/2, 397/2)$ for E_7, $(529, 619)$ for E_8 and $(44, 54)$ for F_4. So every irreducible type other than A_1, A_2, B_2 and G_2 violates (2) on its own, and therefore also inside any product, where κ is larger.

It remains to treat products of A_1, A_2, B_2 and G_2, for which $\min_\gamma q_\gamma$ equals $1/2$, $3/2$, 3 and $15/2$ and κ equals $1/2$, 2, $7/2$ and 8. If some component is A_1, then (2) gives $\kappa \leq 3/2$; every other candidate component has $\kappa \geq 2$, so all components are of type A_1, there are at most three of them, and $\kappa > 1$ leaves A_1^3. If no component is A_1 but one is A_2, then $\kappa \leq 5/2$ and $\Phi = A_2$. In the same way a component B_2 forces $\kappa \leq 4$ and $\Phi = B_2$, and a component G_2 forces $\kappa \leq 17/2$ and $\Phi = G_2$. □

Table 1: The four root types for which the automorphic inequalities reduce, with the central pairing that the spectral cone would need and the barycentric coefficients that prove the reduction.

type	κ	$1/(\kappa-1)$	$(b_{\beta\alpha})$
A_2	2	1	$\begin{pmatrix} 1/3 & 2/3 \\ 2/3 & 1/3 \end{pmatrix}$
A_1^3	3/2	2	$\begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}$
B_2 (α_1 long)	7/2	2/5	$\begin{pmatrix} 1/4 & 2/3 \\ 3/4 & 1/3 \end{pmatrix}$
G_2 (α_1 short)	8	1/7	$\begin{pmatrix} 1/3 & 4/5 \\ 2/3 & 1/5 \end{pmatrix}$

3.2 The matching cones

Corollary 4. Let G be as in §2.1, with root system Φ satisfying $\kappa > 1$ and primitive central pairing d. Then $\mathcal{A} = \mathcal{S}_d$ if and only if $(\Phi, d) = (A_2, 1)$ or $(A_1^3, 2)$.

Proof. By Lemma 2 the cones coincide if and only if Φ is one of the four types of Theorem 3 and $d = 1/(\kappa-1)$. Table 1 gives $1/(\kappa-1) = 1$, 2, $2/5$ and $1/7$, and d is a positive integer by Lemma 1. □

The calibration $d(\kappa-1) = 1$ alone already restricts the root system severely. Let H be the sum of the heights of the positive roots. Since $\langle\beta, \check{\rho}\rangle$ is the height of β, $H = \langle 2\rho, \check{\rho}\rangle = 2\kappa$, so the calibration reads $d(H-2) = 2$ with both factors positive integers, giving $(d, H) = (1, 4)$ or $(2, 3)$. An irreducible component of rank one contributes 1 to H, one of rank two contributes at least 4 (with equality only for A_2), and one of rank $\ell \geq 3$ contributes at least $\ell + 2(\ell-1) \geq 7$, counting the simple roots and one root of height two for each edge of the Dynkin diagram. The calibration therefore allows only A_2 and A_1^4 with $d = 1$ and A_1^3 with $d = 2$. Of these, A_1^4 fails the convex reduction.

Theorem 3 bears on Remark 7.4 of the source. The phenomenon described there, that the inequalities (66) reduce to (58) because one set lies in the convex hull of the other, also occurs for root types B_2 and G_2, for example for GSp_4, $\mathrm{Sp}_4 \times \mathbb{G}_m$ and $G_2 \times \mathbb{G}_m$, none of which is isogenous to the published examples. In those groups the reduction holds and the comparison still fails, because no global form has $d = 2/5$ or $d = 1/7$. If the remark is read as concerning the complete coincidence of the two cones, Corollary 4 proves it at the level of root types and central pairings.

3.3 Two explicit failures

Same Lie algebra, different central pairing. The groups $\mathrm{PGL}_3 \times \mathbb{G}_m$ and GL_3 have isomorphic Lie algebras and the same automorphic cone, but $d = 1$ for the first and $d = 3$ for the second. For GL_3 a cocharacter $\chi = (a, b, c)$ has $x = (a - b, b - c)$ and $z = a + b + c$. The cocharacter $(1, 0, 0)$ has $x = (1, 0)$, $R = 1$ and $z = 1$; it lies in $\mathcal{A}$, while $\mathcal{S}_3$ requires $z = 3$. The cocharacter $(2, 1, 0)$ has $x = (1, 1)$ and $z = 3$; it lies in $\mathcal{S}_3$ and violates $z \leq R/(\kappa - 1) = 2$. Figure 1 shows the two slices at $z = 6$, where $\mathcal{S}_3$ is the slice of $\mathcal{A}$ shrunk by the factor 3. Rescaling θ_G cannot repair this, since $\det^{1/3}$ is not a character of GL_3; a different choice of space or normalization would be a different construction.

Correct central pairing, an extra factor. For $(\mathrm{PGL}_2)^4 \times \mathbb{G}_m$ the root type is A_1^4 with $\kappa = 2$ and $d = 1$, so the calibration holds. Here $R = (x_1 + x_2 + x_3 + x_4)/2$ and $W_\alpha = x_\alpha$. The label $x = (2, 0, 0, 0)$, $z = 1$ has $R = 1$ and lies in $\mathcal{S}_1$, but violates $z \geq W_1 = 2$; so do its three permutations. At $z = 1$ the spectral cone contains 11 integral labels and the automorphic cone 7. Summing the dimensions of the corresponding representations of $\check{G} = \mathrm{SL}_2^4 \times \mathbb{G}_m$ gives 52 on the spectral side and 40 on the automorphic side, the difference being the four omitted three-dimensional representations. Irreducible characters are linearly independent, so no rescaling of either side repairs the comparison term by term.

4. An integrality condition on invariant vectors

Proposition 5.1 of Chen and Venkatesh (2026) states that $k[X] \cong \bigoplus_\lambda V_\lambda^*$, the sum over $\lambda \in X^*(T)$ with $0 \leq \langle \lambda, \check{\rho}' \rangle \leq \min_{\check{\alpha}} \langle \lambda, \check{\alpha} \rangle$. The setup of §5.1 allows $\check{\theta}_G$, and hence $\check{\rho}'$, to be defined only on a finite cover of $\mathbb{G}_m$, and the statement is the same in the arXiv and journal versions. In that generality one more condition is needed.

Proposition 5. Let $\lambda \in X^*(T)$ be dominant. Then $\dim V_\lambda^S \leq 1$, and $V_\lambda^S \neq 0$ if and only if

$$0 \leq \langle \lambda, \check{\rho}' \rangle \leq \min_{\alpha \in \Delta} \langle \lambda, \check{\alpha} \rangle \qquad \text{and} \qquad \langle \lambda, \check{\rho}' \rangle \in \mathbb{Z}.$$

Consequently $k[X] \cong \bigoplus V_\lambda^*$ over the dominant λ satisfying both conditions. The second condition holds for all λ if and only if $\check{\rho}'$ is a cocharacter of T.

Proof. Every weight of V_λ has the form $\mu = \lambda - \sum_\beta k_\beta \beta$ with integers $k_\beta \geq 0$. The central cocharacter $\check{\theta}_G$ pairs trivially with roots and $\langle \beta, \check{\rho} \rangle = 1$ for simple β, so $\langle \mu, \check{\rho}' \rangle = \langle \lambda, \check{\rho}' \rangle - \sum_\beta k_\beta$. Choose N with $N\check{\rho}'$ a cocharacter; then T' is the image of $N\check{\rho}'$, and a weight vector of weight μ is fixed by T' if and only if $\langle \mu, \check{\rho}' \rangle = 0$. If $\langle \lambda, \check{\rho}' \rangle \notin \mathbb{Z}$, no weight of V_λ has this property, so $V_\lambda^{T'} = 0$ and a fortiori $V_\lambda^S = 0$. If $\langle \lambda, \check{\rho}' \rangle \in \mathbb{Z}$, the proof in §5.2 of Chen and Venkatesh (2026) applies unchanged: $V_\lambda^{U'}$ embeds into polynomials in one variable of degree at most $\min_\alpha \langle \lambda, \check{\alpha} \rangle$, and the T'-fixed vectors correspond to multiples of the monomial of degree $\langle \lambda, \check{\rho}' \rangle$. The last statement holds because $X^*(T)$ and $X_*(T)$ are in perfect duality. □

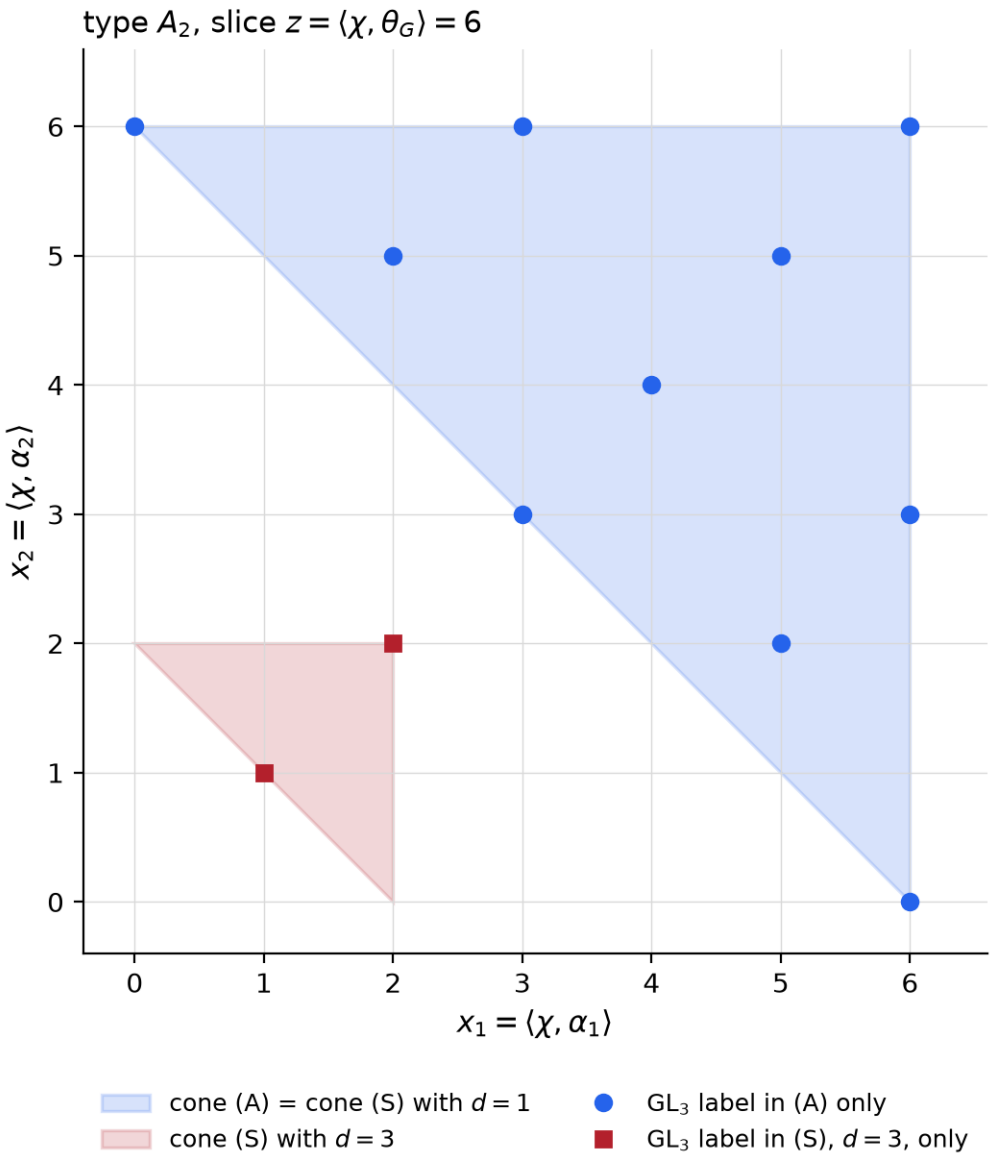


Figure 1: The automorphic cone $\mathcal{A}$ and the spectral cone $\mathcal{S}_d$ for root type A_2, sliced at central degree $z = 6$. For $d = 1$ (the Ginzburg groups $\mathrm{PGL}_3 \times \mathbb{G}_m$ and $\mathrm{SL}_3 \times \mathbb{G}_m$) the two slices coincide. For $d = 3$ (GL_3) the spectral slice is the automorphic slice shrunk by the factor 3, and the two slices are disjoint; the points are the integral labels of GL_3 in each, those with $z \equiv x_1 + 2x_2 \bmod 3$.

The proof in the source already identifies V_λ^S with multiples of $x^{\langle\check\rho',\lambda\rangle}$, which exists as a polynomial only when the exponent is an integer; Proposition 5 makes the condition explicit. The smallest example is $G = \mathrm{SL}_2 \times \mathbb{G}_m$ with θ_G the projection, $\check\theta_G(t) = (1, t)$ and $\check\rho'(s^2) = (\mathrm{diag}(s, s^{-1}), s^2)$. Here U' is trivial and $S = T'$. The standard representation, with trivial central character, has $\langle\lambda, \check\rho'\rangle = 1/2$, so the inequalities $0 \le 1/2 \le 1$ hold, but T' acts on it with weights $s^{\pm1}$ and has no fixed vector.

The automorphic side needs no change. Proposition 5.5 of Chen and Venkatesh (2026) uses Proposition 5.1 only through the real cone spanned by the relevant weights: a linear inequality holds on the relevant weights if and only if it holds on the cone they span. With the integrality condition the relevant weights are the points of a full-rank sublattice inside the same cone, whose positive multiples are still dense in it, so the integrality criterion for points of X and the cone $\mathcal{A}$ are unchanged.

In the four groups of Theorem 7.1 of Chen and Venkatesh (2026) the shifted cocharacter is a genuine cocharacter: $\check\rho'(t) = (\mathrm{diag}(t, 1, t^{-1}), t)$ for $\mathrm{PGL}_3 \times \mathbb{G}_m$ (Chen and Venkatesh 2026, sec. 6.1), $\check\rho$ and $\check\theta$ are cocharacters of $\mathrm{SL}_3 \times \mathbb{G}_m$, the image of $\check\rho'$ for Garrett's group (51) is displayed as the cocharacter $t \mapsto (\mathrm{diag}(t, 1)^{\times 3}, t)$ (Chen and Venkatesh 2026, sec. 6.4), and for the dual group (53) §6.5 of the journal version notes that $\check\rho'$ descends to a genuine cocharacter on the quotient by the diagonal μ_3. As a check of Proposition 5 against independent descriptions of these spaces, the graded dimensions it predicts for $\mathrm{PGL}_3 \times \mathbb{G}_m$ agree through degree 12 with the Hilbert series $(1 - t^2)(1 - t^3)/(1 - t)^8$ of the nilpotent cone of $\mathfrak{sl}_3$, a complete intersection of the two basic invariants (Kostant 1963), and those for Garrett's group agree through degree 12 with the Hilbert series $(1 - t^4)/(1 - t)^8$ of the quartic hypersurface of $(2, 2, 2)$-tensors described in (Chen and Venkatesh 2026, sec. 6.4).

5. When equal cones give equal supports

5.1 The criterion

The first identity of the duality compares the automorphic X-period of a cusp form on G with the spectral $\check X$-period of its parameter. At a place v the automorphic local factor is indexed by

$$\mathcal{A}_\Lambda = \{\chi \in X_*(T) \text{ dominant} : \ \chi \in \mathcal{A}\},$$

and by Proposition 5 applied to $\check G$, after $\lambda \mapsto -w_0\lambda$, the representations occurring in $k[\check X]$ have highest weights in

$$\mathcal{S}_\Lambda = \{\chi \in X^*(\check T) : \ \chi \in \mathcal{S}_d, \ \langle\chi, \check\rho_{\check G} - \check\theta_{\check G}\rangle \in \mathbb{Z}\}.$$

Both are subsets of the lattice $\Lambda = X_*(T) = X^*(\check T)$. In the coordinates of §2.2 the integrality condition reads $R(x) - z/d \in \mathbb{Z}$. If $\check G \times \mathbb{G}_{gr}$ acts with $\mathbb{G}_{gr}$ through $d\,\check\theta_{\check G}$, the grading of V_χ is $\langle\chi, d\check\theta_{\check G}\rangle = \langle\chi, \theta_G\rangle = z$, the same exponent that weights s_χ on the automorphic side.

Theorem 6. Suppose $\mathcal{A} = \mathcal{S}_d$. Then $\mathcal{A}_\Lambda = \mathcal{S}_\Lambda$ if and only if $\check\rho'_{\check G}$ is a cocharacter of $\check T$, equivalently $\rho - \theta_G/d \in X^*(T)$. In that case the local factors $\sum_\chi q_v^{-z/2} s_\chi$ on the two sides coincide as functions of the Satake parameter. Otherwise they differ, already as formal sums of irreducible characters.

The same holds for the second identity, with G and $\check{G}$ exchanged, under the condition that $\check{\rho}'_G$ is a cocharacter of T; the cones for the exchanged pair coincide as well, because $\check{\Phi}$ has the same type as Φ in the two cases of Corollary 4 and $d_{\check{G}} = d_G$.

Proof. Under the hypothesis, $\mathcal{A}_\Lambda$ consists of all points of Λ in the cone and $\mathcal{S}_\Lambda$ of those on which the functional $\phi(\chi) = \langle \chi, \check{\rho}_{\check{G}} - \check{\theta}_{\check{G}} \rangle$ is an integer. The long Weyl element maps $\check{\rho}_{\check{G}} + \check{\theta}_{\check{G}}$ to $-(\check{\rho}_{\check{G}} - \check{\theta}_{\check{G}})$ and preserves cocharacters, so ϕ is integral on Λ if and only if $\check{\rho}'_{\check{G}}$ is a cocharacter. If it is, the two sets agree. If it is not, some $\mu \in \Lambda$ has $\phi(\mu) \notin \mathbb{Z}$. The cone is full-dimensional, so it contains a point $p \in \Lambda$ in its interior, and $Np + \mu$ lies in the cone for large N. One of Np and $Np + \mu$ has non-integral ϕ, and it lies in $\mathcal{A}_\Lambda$ but not in $\mathcal{S}_\Lambda$. The statement about local factors follows from the linear independence of irreducible characters. □

The theorem concerns only the term-by-term comparison of local factors that the proof of Theorem 7.1 of Chen and Venkatesh (2026) performs; it does not exclude a global identity obtained in another way.

5.2 Root type A_2

For $(A_2, d = 1)$ the element θ_G pairs to 1 with ν_G, so θ_G itself is primitive in the character lattice, and the lattice splits as $L' \times \mathbb{Z}$ with L' between the root lattice and the weight lattice of A_2. There are two global forms, $\mathrm{PGL}_3 \times \mathbb{G}_m$ and $\mathrm{SL}_3 \times \mathbb{G}_m$, dual to each other. In both, $\langle \lambda, \check{\rho}' \rangle = p_1 + p_2 + r$ in terms of $p_\alpha = \langle \lambda, \check{\alpha} \rangle$ and $r = \langle \lambda, \nu_G \rangle$, because $m_1 = m_2 = 1$ and $d = 1$, so the integrality condition holds automatically and both support comparisons go through.

5.3 Root type A_1^3

For root type A_1^3 with $d = 2$, record a character λ by $(p_1, p_2, p_3, r) \in \mathbb{Z}^4$ with $p_i = \langle \lambda, \check{\alpha}_i \rangle$ and $r = \langle \lambda, \nu_G \rangle$. The roots are $2e_1, 2e_2, 2e_3$ and $\theta_G = 2e_4$, so the character lattice L satisfies $2\mathbb{Z}^4 \subseteq L \subseteq \mathbb{Z}^4$ and is determined by the binary code $C = L/2\mathbb{Z}^4 \subseteq \mathbb{F}_2^4$. Conversely every subspace C defines a root datum of type A_1^3 with one-dimensional centre, and the primitivity of θ_G and of ν_G means that $e_4 \notin C$ and that some word of C has last coordinate 1. Call such codes admissible. Write a rational cocharacter as $v = \sum_{i \le 3} v_i \check{\alpha}_i + v_4 \nu_G$. It lies in $X_*(T)$ if and only if $\sum_i v_i \ell_i \in \mathbb{Z}$ for every $\ell \in L$, that is, if and only if $2v$ is an integral vector orthogonal to C modulo 2. Its coordinates as a character of $\check{T}$, obtained by pairing with the coroots α_i of $\check{G}$ and with $\nu_{\check{G}} = \theta_G$, are $2v$; so the dual group has code $C^\perp$ for the standard dot product on $\mathbb{F}_2^4$. Finally $\langle \lambda, \check{\rho}'_G \rangle = (p_1 + p_2 + p_3 + r)/2$, since $m_i = 1/2$ and $\check{\theta}_G = \nu_G/2$.

Proposition 7. Let $u = (1, 1, 1, 1)$. Among the 67 subspaces of $\mathbb{F}_2^4$, 35 are admissible. For an admissible code C, $\check{\rho}'_G$ is a cocharacter if and only if $C \subseteq u^\perp$, and $\check{\rho}'_{\check{G}}$ is a cocharacter if and only if $u \in C$. Both hold for exactly five codes, the subspaces with $\langle u \rangle \subseteq C \subseteq u^\perp$; exactly one holds for 12 codes; neither holds for 18.

Proof. The subspaces of $\mathbb{F}_2^4$ number $1 + 15 + 35 + 15 + 1 = 67$. Those inside $e_4^\perp$ correspond to subspaces of $\mathbb{F}_2^3$ and those containing e_4 to subspaces of $\mathbb{F}_2^4/\langle e_4 \rangle$, 16 each, and no subspace has

both properties because $e_4 \notin e_4^{\perp}$; this leaves $67 - 32 = 35$. The first criterion is the formula for $\langle \lambda, \check{\rho}'_G \rangle$; the second is the first applied to $C^{\perp}$, since $C^{\perp} \subseteq u^{\perp}$ if and only if $u \in C$. Subspaces between $\langle u \rangle$ and $u^{\perp}$ correspond to subspaces of the plane $u^{\perp}/\langle u \rangle$, of which there are five, and they are admissible because u has last coordinate 1 and $e_4 \notin u^{\perp}$. The counts 12 and 18 are by enumeration. □

Table 2: Global forms of root type A_1^3 with $d = 2$; (51) and (53) are Garrett's groups as numbered by Chen and Venkatesh (2026, secs. 6.4–6.5).

group	code C	$\check{\rho}'_G$ integral	$\check{\rho}'_{\check{G}}$ integral
Garrett's group (51)	$\langle u \rangle$	yes	yes
its dual (53)	$u^{\perp}$	yes	yes
$\mathrm{SO}_4 \times \mathrm{GL}_2$ (three labellings)	$\langle u, e_1 + e_2 \rangle$ and permutations	yes	yes
$\mathrm{GL}_2 \times \mathrm{PGL}_2 \times \mathrm{PGL}_2$	$\langle e_1 + e_4 \rangle$	yes	no
$\mathrm{GL}_2 \times \mathrm{SL}_2 \times \mathrm{SL}_2$	$\langle e_1 + e_4, e_2, e_3 \rangle$	no	yes

The codes in Table 2 follow from explicit characters: the tensor representation of Garrett's group has $(p, r) = (1, 1, 1, 1)$; for $\mathrm{SO}_4 = (\mathrm{SL}_2 \times \mathrm{SL}_2)/\mu_2$ the first two labels have equal parity; for GL_2 the label and the central weight have equal parity. The five codes of Proposition 7 consist of Garrett's pair, the line $\langle u \rangle$ and the plane $u^{\perp}$, which are dual to each other, and three self-dual codes, which differ by a permutation of the three factors and all give the group $\mathrm{SO}_4 \times \mathrm{GL}_2$. This group is its own Langlands dual, and Corollary 4 and Theorem 6 show that the support comparison in §7.4 of Chen and Venkatesh (2026) holds for it in both directions, with the same grading and the same exponent of the discriminant: $\dim G/U - (2\kappa - 1)d = 7 - 4 = 3$, as in (68) of the source. We have not checked the other inputs of the proof of Theorem 7.1 for this group, in particular the condition of §7.3 that every smaller orbit on X have a stabilizer containing the unipotent radical of a proper parabolic subgroup, so we make no claim about its periods.

5.4 An isogenous form with one failing comparison

The group $G = \mathrm{GL}_2 \times \mathrm{PGL}_2 \times \mathrm{PGL}_2$ has dual $\check{G} = \mathrm{GL}_2 \times \mathrm{SL}_2 \times \mathrm{SL}_2$, root type A_1^3 and $d = 2$, so its cones coincide with those of Garrett's example. Its own shifted cocharacter is integral, while that of $\check{G}$ is not. For the first identity, $\Lambda = X_*(T)$ consists, in the coordinates of §2.2 with $p_i = x_i$, of the (p_1, p_2, p_3, z) with $p_1 \equiv z \pmod 2$, and the integrality condition $(p_1 + p_2 + p_3 - z)/2 \in \mathbb{Z}$ removes the labels with $p_2 + p_3$ odd. The two supports agree in central degrees $z \leq 2$. At $z = 3$ the automorphic support has 12 labels and the spectral support 10; the missing labels are $(1, 1, 2)$ and $(1, 2, 1)$, each of dimension 12 for $\check{G}$, so the summed dimensions are 214 and 190. At $z = 8$ the counts are 125 and 85 (Figure 2). By Proposition 5 the label $(p; z) = (1, 1, 2; 3)$, for which the invariant would have to be a polynomial of degree $1/2$, gives no S-invariant vector: the torus T'

acts on the 12-dimensional representation with odd weights only. For the second identity, with $\check{G}$ on the automorphic side, the two supports agree in every degree, as Theorem 6 predicts; the computation confirms this through $z = 8$.

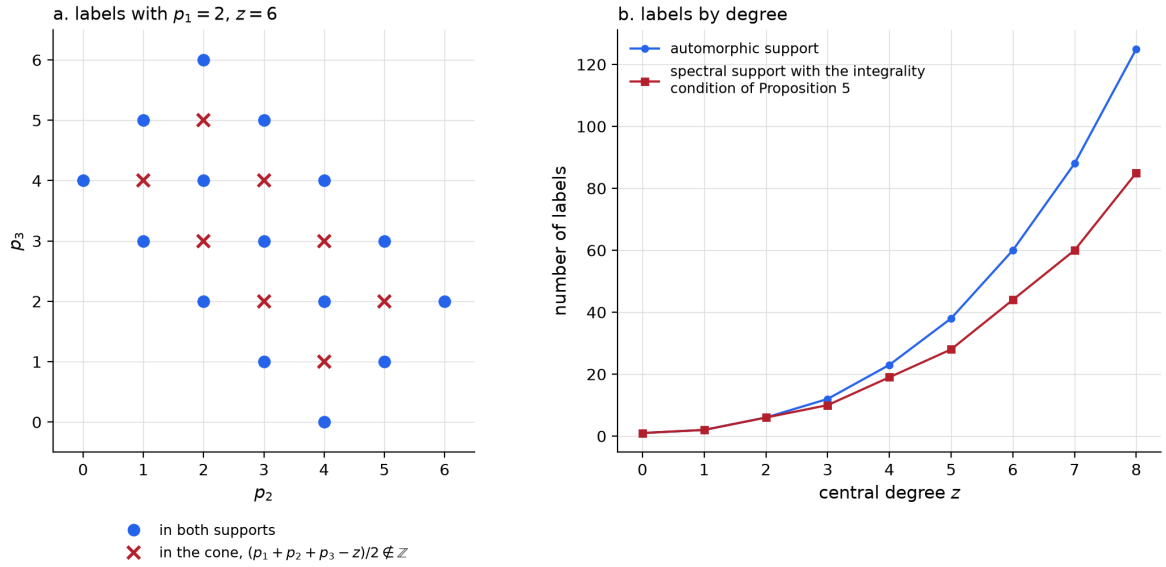


Figure 2: Integrality for $G = \mathrm{GL}_2 \times \mathrm{PGL}_2 \times \mathrm{PGL}_2$ and $\check{G} = \mathrm{GL}_2 \times \mathrm{SL}_2 \times \mathrm{SL}_2$, whose cones coincide with those of Garrett's example. (a) Labels with $p_1 = 2$ at central degree $z = 6$ in the common cone; crosses mark labels of the automorphic support that the integrality condition of Proposition 5 removes from the spectral support. (b) Number of labels in each support by central degree.

6. Exact checks

The finite claims above were checked by a program in exact rational arithmetic, recorded with the paper. It computes the Cartan data, positive roots, n, m and κ for all 470 semisimple types of rank at most 8 with $\kappa > 1$, checking n and m against half-sums of positive roots and coroots and 2κ against H. For each type it decides (1) by linear programming and confirms the answer with an exact certificate: a rational point violating (1), or nonnegative barycentric coefficients. The reduction holds for A_2, B_2, G_2 and A_1^3 and for no other type, and the integral calibration only for A_2, A_1^3 and A_1^4, in agreement with Theorem 3 and Corollary 4. It also verifies the closed forms of q_1 and κ used in the proof for all classical types of rank at most 8, and compares the cones directly on integral boxes: 1,377 points for $(A_2, 1)$ and 3,456 for $(A_1^3, 2)$ with no disagreement, and disagreement for $(A_2, 3)$ and $(A_1^4, 1)$.

Proposition 5 was tested by computing $\dim V_\lambda^S$ directly. For root types A_1, A_1^2 and A_1^3, V_λ was realized as a tensor product of symmetric powers with the raising operators acting on monomials, and V_λ^S computed as the kernel of $\mathrm{Lie}(U')$ on the zero-weight space of T', for 799 pairs of a highest weight and a half-integral central weight. The dimension was at most one, and equal to one exactly when both conditions of Proposition 5 hold; in 29 of the 105 cases satisfying the inequalities, the integrality condition fails and there is no invariant. For $\mathrm{SL}_3 \times \mathbb{G}_m$, $V_{(a,b)}$ was realized

as the harmonic polynomials of bidegree (a, b) in x_1, x_2, x_3 and y_1, y_2, y_3, and all 144 cases with $a, b \leq 3$ agree with the criterion. The program enumerates the 67 subspaces of $\mathbb{F}_2^4$ for Proposition 7, computes the supports of §5.4 up to $z = 8$ and the Hilbert series of §4 up to degree 12, and runs 26 checks in all. These computations corroborate the proofs; the infinite statements rest on the proofs.

7. What the comparison depends on

The comparison of supports in the singular examples rests on three conditions of different kinds. The convex reduction depends only on the root system, and it holds for four types. The calibration $d(\kappa-1) = 1$ couples the root system to the way the centre meets the derived group, and it removes B_2 and G_2. The integrality of the shifted cocharacters depends on the full character lattice, and among the 35 labelled global forms of type A_1^3 with $d = 2$ it keeps five. The published examples satisfy all three at once. By Corollary 4 the uniform construction can succeed only in their two root types, and by Proposition 7 only for particular lattices within them.

For the global forms that fail the integrality condition, the naive space $\check{X} = \overline{\check{T}'\check{U}'\backslash\check{G}}$ has too few functions. For tori, Chen proposes provisional definitions in which a space with finite stabilizers is paired with a Deligne–Mumford quotient stack, proves weak numerical duality for such pairs, and expects the same for central isogenies of reductive groups (Chen 2024, sec. 6 and Theorem 6.10); a stacky spectral side also appears in the GSO_6 example of Cauchi, Chen and Gutierrez Terradillos (Cauchi et al. 2026). Theorem 6 identifies the forms in the A_2 and A_1^3 cases for which no such modification is needed, and Figure 2 shows the size of the discrepancy for one form that needs it. Whether a stacky $\check{X}$ restores the term-by-term matching for $\mathrm{GL}_2 \times \mathrm{PGL}_2 \times \mathrm{PGL}_2$ is a natural test case.

There is a structural analogy on the physical side of the duality. In four-dimensional gauge theory, one Lie algebra admits several global forms of the gauge group, which are distinguished by their line operators and share the correlation functions of local operators (Aharony et al. 2013). The comparison studied here likewise distinguishes groups with the same Lie algebra, and the binary codes of Proposition 7 resemble the finite data that label global forms. The analogy is not an identification: no argument given here matches the arithmetic integrality conditions with line operators. Mathematical constructions of the dual Hamiltonian space, such as Nakajima's in the cotangent-type case (Nakajima 2024), would be the place to make such a comparison precise.

Three questions remain open. Do the remaining inputs of the proof of Theorem 7.1 of Chen and Venkatesh (2026), above all the orbit condition of §7.3, hold for $\mathrm{SO}_4 \times \mathrm{GL}_2$, so that $\overline{T'U'\backslash(\mathrm{SO}_4 \times \mathrm{GL}_2)}$ is numerically weakly self-dual with discrepancy 3? For the 12 forms of type A_1^3 that pass one integrality test, does the identity whose supports agree hold? And for B_2 and G_2, where the convex reduction holds but no global form has the required central pairing, is there a modification of the spectral space or of the grading that produces a matching cone? Other singular constructions, such as the GSO_6 example of Cauchi et al. (2026), lie outside the recipe $\overline{T'U'\backslash G}$ and outside the scope of these results.

The program and a record of the run that produced every number and figure are in the repository; `simulation/run_all.py` reproduces them.

References

Aharony, Ofer, Nathan Seiberg, and Yuji Tachikawa. 2013. "Reading Between the Lines of Four-Dimensional Gauge Theories." *Journal of High Energy Physics* 2013 (8): 115. https://doi.org/10.1007/jhep08(2013)115.

Ben-Zvi, David, Yiannis Sakellaridis, and Akshay Venkatesh. 2024. "Relative Langlands Duality." 2409.04677. Preprint, arXiv. https://doi.org/10.48550/arXiv.2409.04677.

Beuzart-Plessis, Raphaël. 2025. "Introduction to the Relative Langlands Program." 2509.18062. Preprint, arXiv. https://doi.org/10.48550/arXiv.2509.18062.

Borel, A. 1979. "Automorphic L-Functions." In *Automorphic Forms, Representations and L-Functions, Part 2*, edited by A. Borel and W. Casselman, vol. 33. Proceedings of Symposia in Pure Mathematics. American Mathematical Society. https://doi.org/10.1090/pspum/033.2/546608.

Bourbaki, Nicolas. (1968) 2002. *Lie Groups and Lie Algebras, Chapters 4–6*. Translated by Andrew Pressley. Elements of Mathematics. Springer. https://doi.org/10.1007/978-3-540-89394-3.

Bouthier, A., B. C. Ngô, and Y. Sakellaridis. 2016. "On the Formal Arc Space of a Reductive Monoid." *American Journal of Mathematics* 138 (1): 81–108. https://doi.org/10.1353/ajm.2016.0004.

Braverman, Alexander, and David Kazhdan. 1999. "On the Schwartz Space of the Basic Affine Space." *Selecta Mathematica (New Series)* 5 (1): 1–28. https://doi.org/10.1007/s000290050041.

Cauchi, Antonio, Eric Yen-Yo Chen, and Armando Gutierrez Terradillos. 2026. "Relative Langlands Duality of the Bump–Friedberg–Ginzburg GSO_6-Integral." 2608.30576. Preprint, arXiv. https://doi.org/10.48550/arXiv.2608.30576.

Chen, Eric Y. 2024. "Relative Langlands Duality of Toric Periods." 2405.18231. Preprint, arXiv. https://doi.org/10.48550/arXiv.2405.18231.

Chen, Eric Y., and Akshay Venkatesh. 2026. "Some Singular Examples of Relative Langlands Duality." *Selecta Mathematica (New Series)* 32 (5): 97. https://doi.org/10.1007/s00029-026-01202-5.

Feng, Tony, and Jonathan Wang. 2025. "Geometric Langlands Duality for Periods." *Geometric and Functional Analysis* 35 (2): 463–541. https://doi.org/10.1007/s00039-025-00702-4.

Gaiotto, David, and Edward Witten. 2009. "S-Duality of Boundary Conditions in $\mathcal{N} = 4$ Super Yang–Mills Theory." *Advances in Theoretical and Mathematical Physics* 13 (3): 721–896. https://doi.org/10.4310/atmp.2009.v13.n3.a5.

Garrett, Paul B. 1987. "Decomposition of Eisenstein Series: Rankin Triple Products." *Annals of Mathematics* 125 (2): 209–35. https://doi.org/10.2307/1971310.

Ginzburg, David. 1991. "A Rankin–Selberg Integral for the Adjoint Representation of GL_3." *Inventiones Mathematicae* 105 (1): 571–88. https://doi.org/10.1007/bf01232279.

Humphreys, James E. 1972. *Introduction to Lie Algebras and Representation Theory*. Graduate Texts in Mathematics 9. Springer. https://doi.org/10.1007/978-1-4612-6398-2.

Kapustin, Anton, and Edward Witten. 2007. "Electric-Magnetic Duality and the Geometric Langlands Program." *Communications in Number Theory and Physics* 1 (1): 1–236. https://doi.org/10.4310/cntp.2007.v1.n1.a1.

Knop, Friedrich, and Barbara Schalke. 2017. "The Dual Group of a Spherical Variety." *Transactions of the Moscow Mathematical Society* 78: 187–216. https://doi.org/10.1090/mosc/270.

Kostant, Bertram. 1963. "Lie Group Representations on Polynomial Rings." *American Journal of Mathematics* 85 (3): 327–404. https://doi.org/10.2307/2373130.

Nakajima, Hiraku. 2024. "S-Dual of Hamiltonian G Spaces and Relative Langlands Duality." 2409.06303. Preprint, arXiv. https://doi.org/10.48550/arXiv.2409.06303.

Sakellaridis, Yiannis. 2012. "Spherical Varieties and Integral Representations of L-Functions." *Algebra & Number Theory* 6 (4): 611–67. https://doi.org/10.2140/ant.2012.6.611.

Sakellaridis, Yiannis, and Akshay Venkatesh. 2017. "Periods and Harmonic Analysis on Spherical Varieties." *Astérisque* 396. https://doi.org/10.24033/ast.1040.

Sakellaridis, Yiannis, and Jonathan Wang. 2022. "Intersection Complexes and Unramified L-Factors." *Journal of the American Mathematical Society* 35 (3): 799–910. https://doi.org/10.1090/jams/990.

Springer, T. A. 1998. *Linear Algebraic Groups*. 2nd ed. Progress in Mathematics 9. Birkhäuser. https://doi.org/10.1007/978-0-8176-4840-4.